

ROHAT GUNES

SENIOR SOFTWARE DEVELOPER

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WORK EXPERIENCE

Senior Software Developer

[Ericsson AB](#) 

 June 2021 – Present

 Lund, Sweden

- Ericsson proprietary algorithm development on **ISAC** specified in [3GPP TS 22.916](#) .
- **Time-complexity analysis** and cost predictions using ML models, **overhaul of entire ML pipeline**.
- Software systemization, design and implementation on control interfaces and components of L1.
- Fixed point implementations of Ericsson proprietary L1 algorithms in **PUSCH, SRS**.
- **Python, C and C++ 17** development in **Linux** environment for various baseband related tasks.
- Multicore DSP Programming within **Ericsson Many Core Architecture**.

Software Developer

[Ulak Comms Inc.](#) 

 May 2019 – May 2021

 Istanbul, Turkey

- Software design and implementation using **C++14** to achieve an optimized **L2RRM**, scheduler algorithm.
- Implemented **multicore DSP programming** modules on TI c66x chipsets.
- MATLAB simulations for over the air timing offset estimations in **NB-IoT** scenarios, e.g. EPA 5Hz propagation model.
- Floating point simulations of physical channel estimation and equalization processes in **PUSCH**.
- Bit exactness verification between floating and fixed point implementations.
- Time critical software implementation in **PUSCH** to deploy **UL CoMP** technology by leveraging **LLR Combining**.
- Software implementation of both control and user plane in **RLC/PDCP** layers for **NB-IoT**.
- Implementation of Packet Delay Measurement Metrics for L2 as specified in **3GPP TS 36.314**.

Part-time Software Developer

[Vestel](#) 

 Dec 2017 – May 2018

 Manisa, Turkey

- Improving security protocols and algorithms used for wireless communications in smarthome devices.

EDUCATION

BSc in Electrical and Electronics Eng.

[Ege University](#) 

 Sep 2014 – Jul 2018

 Izmir, Turkey

- Telecommunications Elective Courses - 240 ECTS in total - GPA 3.29/4.00
- **Graduation Project: USB Powered Ultra Wideband Superheterodyne Receiver Design**
Implemented a receiver system in 1-4.2 GHz band. Achieving a minimum gain of 13 dB and noise figure below 3 dB. The produced RF components were measured with commercial network analyzers. The software interface was built within LabView.

ACHIEVEMENTS AND SCHOLARSHIPS

Coursera Certificate: [Object-Oriented Data Structures in C++](#) 

Scholarship: Granted by the [Ministry of Youth and Sports](#)  upon achieving high rankings in national university entrance exams.

LANGUAGES

Turkish: Native

English: Fluent, *TOEFL IBT 100/120 (July 2019)*

Swedish: Elementary, *completed SFI D-Course (Juni 2022)*

SKILLS & INTERESTS

Skills: C/C++, Python, Bazel, Make, signal processing, machine learning, data visualization, data collection, system design, public speech, ...

Interests: Hiking, climbing, gym, snowboarding (inactive), board games

Music: post metal/rock, Godspeed You! Black Emperor, Year of No Light